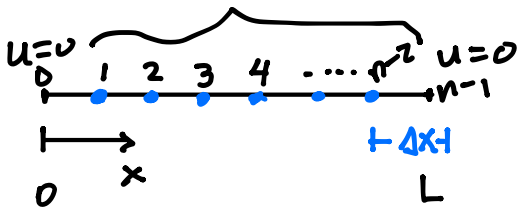
$u(x,t)$



n – nodos

$$\frac{\partial^2 u}{\partial x^2} = c\,\frac{\partial v}{\partial t}$$

$t=0 \quad u(x,0) = u_i = cte$

$$\begin{cases} j = 1,2,3 \dots n-2 \\ \dfrac{u_{j+1} - 2v_j + v_{j-1}}{\Delta x^2} = c\,\dfrac{dv_j}{dt} \end{cases}$$

$$\left.\begin{matrix} j=0 \\ u_j = 0 \end{matrix}\right\} v_j^{k+1} = 0$$

$$\left.\begin{matrix} j = n-1 \\ u_j = 0 \end{matrix}\right\} v_j^{k+1} = 0$$

Explicito

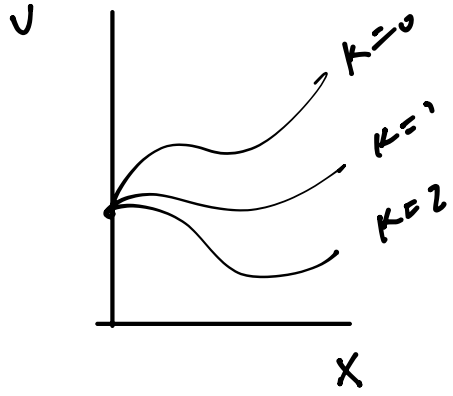
$$c\,\frac{v_j^{k+1} - v_j^k}{\Delta t} = \frac{v_{j+1}^k - 2v_j^k + v_{j-1}^k}{\Delta x^2}$$

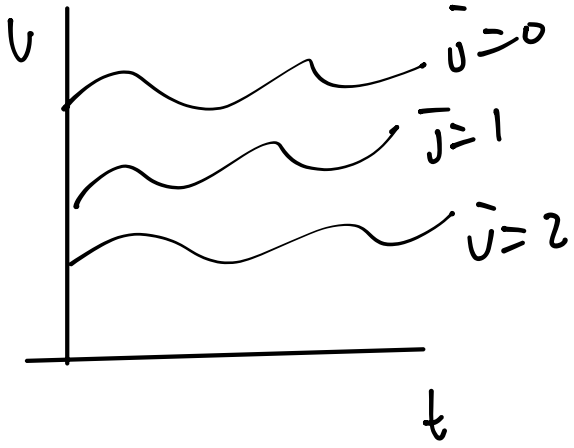
$$v_j^{k+1} = \frac{\Delta t}{c}\left(\frac{v_{j+1}^k - 2v_j^k + v_{j-1}^k}{\Delta x^2}\right) + v_j^k$$

$$\begin{bmatrix} v_0^0 = v_i = cte \\ v_1^0 \\ v_2^0 \\ \vdots \\ v_{n-1}^0 \end{bmatrix} \xrightarrow{\Delta t} \begin{bmatrix} v_0^1 \\ v_1^1 \\ v_2^1 \\ \vdots \\ v_{n-1}^1 \end{bmatrix} \xrightarrow{\Delta t} \begin{bmatrix} v_0^2 \\ v_1^2 \\ \vdots \\ v_{n-1}^2 \end{bmatrix} \dots$$

$k=0 \quad k=1 \quad k=2$

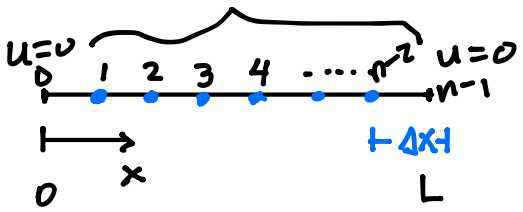
$$H = \begin{bmatrix} U_0^0 & U_0^1 & U_0^2 & \cdots \\ \vdots & \ddots & \vdots & \\ U_{n-1}^0 & U_{n-1}^1 & U_{n-1}^2 & \end{bmatrix} \begin{matrix} \leftarrow \bar{U}=0 \\ \leftarrow \bar{U}=1 \\ \leftarrow \bar{U}=2 \\ \vdots \end{matrix}$$





$U(x,t)$

$u(x,t)$



n – nodo

$$\frac{\partial^2 u}{\partial x^2} = c\,\frac{\partial v}{\partial t}$$

$t=0 \quad u(x,0) = u_i = cte$

$$\begin{cases} j = 1,2,3\ldots n-2 \\ \dfrac{U_{j+1}^{k+1} - 2U_j^{k+1} + U_{j-1}^{k+1}}{\Delta x^2} = c\,\dfrac{U_j^{k+1} - U_j^{k}}{\Delta t} \end{cases}$$

$j=0 \quad U_j^{k+1} = 0$ $\qquad$ $j=n-1 \quad U_j^{k+1} = 0$

$j=0 \quad A_{j,j} = 1 \quad B_j = 0$

$j=1\ldots n-2 \quad A_{j,j+1} = \dfrac{1}{\Delta x^2}$

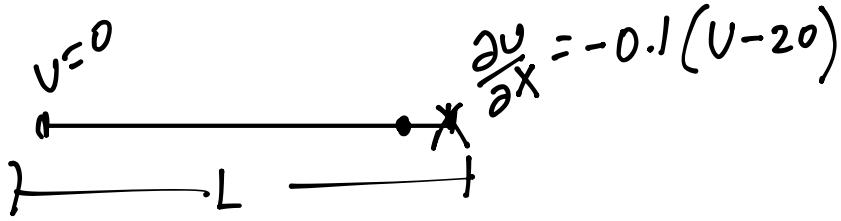
$$B_j = -\frac{c}{\Delta t}\,U_j^k$$

$$A_{j,j} = \frac{-2}{\Delta x^2} - \frac{c}{\Delta t}$$

$$A_{j,j-1} = \frac{1}{\Delta x^2}$$

$j=n-1 \quad A_{j,j} = 1 \quad B_j = 0$

$$\frac{\partial^2 v}{\partial x^2} = c\frac{\partial v}{\partial t}$$



$$\frac{\partial v}{\partial x} \approx \frac{v_j^{k+1} - v_{j-1}^{k+1}}{\Delta x} = -0.1\left(v_j^{k+1} - 20\right)$$

$$A_{j,j} = \frac{1}{\Delta x} + 0.1 \qquad A_{j,j-1} = \frac{-1}{\Delta x} \qquad B_j = (-0.1)(-20)$$

$$\frac{\partial^2 v}{\partial x^2} = c\frac{\partial v}{\partial t} + 0.1x^2$$